

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12401934
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Huang; Ming-Tau et al.

Semiconductor device package and acoustic device having the same

Abstract

A semiconductor device package comprises a first substrate, a first electronic component, an encapsulant, and a feeding structure. The first substrate has a first surface and a second surface opposite the first surface. The first electronic component is disposed on the first surface of the first substrate. The encapsulant encapsulates the first electronic component on the first surface of the first substrate. The feeding structure is disposed on the second surface of the first substrate without covering.

Inventors: Huang; Ming-Tau (Kaohsiung, TW), Tseng; Kuei-Hao (Kaohsiung, TW), Lin; Hung-I (Kaohsiung, TW)

Applicant: Advanced Semiconductor Engineering, Inc. (Kaohsiung, TW)

Family ID: 1000008778522

Assignee: ADVANCED SEMICONDUCTOR ENGINEERING, INC. (Kaohsiung, TW)

Appl. No.: 17/968725

Filed: October 18, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230039552 A1	Feb. 09, 2023

Related U.S. Application Data

continuation parent-doc US 16528338 20190731 US 11477559 child-doc US 17968725

Publication Classification

Int. Cl.: H04R1/10 (20060101); B81B7/00 (20060101)

U.S. Cl.:

CPC **H04R1/1016** (20130101); **B81B7/008** (20130101); H04R2201/003 (20130101)

Field of Classification Search

CPC: H04R (1/1016); H04R (2201/003); B81B (7/008)

USPC: 381/74; 381/58; 381/151; 381/311; 381/380; 381/374; 381/150; 438/106; 438/107;
 438/112; 343/791; 343/866; 455/572; 320/127; 320/128; 320/137

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7570218	12/2008	Tsujimura et al.	N/A	N/A
8315060	12/2011	Morikita et al.	N/A	N/A
9118990	12/2014	Hankey et al.	N/A	N/A
9932221	12/2017	Natan et al.	N/A	N/A
10049893	12/2017	Chen et al.	N/A	N/A
11477559	12/2021	Huang et al.	N/A	N/A
2003/0169209	12/2002	Ohara et al.	N/A	N/A
2004/0084771	12/2003	Bolken et al.	N/A	N/A
2007/0029667	12/2006	Fujii et al.	N/A	N/A
2007/0080360	12/2006	Mirsky et al.	N/A	N/A
2008/0164545	12/2007	Hsiao	N/A	N/A
2012/0049348	12/2011	Ro	N/A	N/A
2014/0198927	12/2013	Cheng	N/A	N/A
2015/0350762	12/2014	Birger et al.	N/A	N/A
2016/0050474	12/2015	Rye et al.	N/A	N/A
2016/0073189	12/2015	Linden et al.	N/A	N/A
2016/0225745	12/2015	Beer et al.	N/A	N/A
2017/0195767	12/2016	Kim et al.	N/A	N/A
2017/0209095	12/2016	Wagner et al.	N/A	N/A
2017/0330870	12/2016	Chen et al.	N/A	N/A
2018/0323170	12/2017	Kim et al.	N/A	N/A
2019/0045294	12/2018	Kim et al.	N/A	N/A
2020/0221210	12/2019	Grebe	N/A	H01Q 1/40

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
107623881	12/2017	CN	N/A
208754499	12/2018	CN	N/A
2001-217332	12/2000	JP	N/A
2001-352212	12/2000	JP	N/A
2007-042978	12/2006	JP	N/A
2007-288360	12/2006	JP	N/A
2010-516096	12/2009	JP	N/A
2011-097306	12/2010	JP	N/A
2012-074505	12/2011	JP	N/A

2017-204635	12/2016	JP	N/A
2021-027308	12/2020	JP	N/A
WO-2005/067359	12/2004	WO	N/A
WO-2009/122835	12/2008	WO	N/A
WO-2015/068403	12/2014	WO	N/A

OTHER PUBLICATIONS

First Office Action, with English summary for counterpart Japanese Patent Application No. 2019-198831, issued Oct. 31, 2023, 9 pages. cited by applicant

Extended Search Report for corresponding European Patent Application No. 19204047.5, issued on Jun. 5, 2020, 7 pages. cited by applicant

Final Office Action U.S. Appl. No. 16/528,338, issued Nov. 10, 2020, 39 pages. cited by applicant

Final Office Action U.S. Appl. No. 16/528,338, issued, Sep. 14, 2021, 46 pages. cited by applicant

Non-Final Office Action for U.S. Appl. No. 16/528,338, issued May 29, 2020, 20 pages. cited by applicant

Non-Final Office Action U.S. Appl. No. 16/528,338, issued Feb. 3, 2022, 16 pages. cited by applicant

Non-Final Office Action U.S. Appl. No. 16/528,338, issued Mar. 22, 2021, 37 pages. cited by applicant

Notice of Allowance U.S. Appl. No. 16/528,338, issued Jun. 13, 2022, 9 pages. cited by applicant

Office Action for corresponding European Patent Application No. 19204047.5, issued on Jun. 21, 2021, 6 pages. cited by applicant

Office Action, for counterpart Chinese Patent Application No. 201910764286.3, issued Nov. 28, 2024, 7 pages. cited by applicant

Search Report with English translation, counterpart Chinese Patent Application No. 201910764286.3, issued Nov. 28, 2024, 4 pages. cited by applicant

Office Action, for counterpart Japanese Patent Application No. 2024-063839, issued Mar. 5, 2025, 5 pages. cited by applicant

Summary with English Translation of Office Action, for counterpart Japanese Patent Application No. 2024-063839, issued Mar. 5, 2025, 1 page. cited by applicant

Office Action for counterpart European Patent Application No. 19204047.5, issued Jan. 24, 2025, 8 pages. cited by applicant

Search Report with English translation for Chinese Application No. 201910764286.3 dated Jul. 7, 2025, 4 pages. cited by applicant

Primary Examiner: Chin; Vivian C

Assistant Examiner: Tran; Con P

Attorney, Agent or Firm: FOLEY & LARDNER LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 16/528,338 filed Jul. 31, 2019, now issued as U.S. Pat. No. 11,477,559, the contents of which is incorporated herein by reference in its entirety.

BACKGROUND

1. Technical Field

(1) The present disclosure relates generally to a semiconductor device package, and more particularly, the present disclosure relates to an acoustic device having the semiconductor device package.

2. Description of the Related Art

(2) As technology advances, various circuits or modules can be integrated into one electronic device, e.g. an acoustic device (e.g. an earphone or a wireless earphone) to perform multifunction. However, miniaturization of the electronic device may adversely affect performance of the electronic device.

SUMMARY

(3) In some embodiments, the present disclosure provides a semiconductor device package. The semiconductor device package includes a first substrate, a first electronic component, an encapsulant and a feeding structure. The substrate has a first surface and a second surface opposite the first surface. The first electronic component is disposed on the first surface of the first substrate. The encapsulant encapsulates the first electronic component on the first surface of the first substrate. The feeding structure is disposed on the second surface of the first substrate without covering.

(4) In some embodiments, the present disclosure further provides a wireless earphone. The wireless earphone includes a semiconductor device package. The semiconductor device package includes a first substrate, a first electronic component, an encapsulant and a feeding structure. The substrate has a first surface and a second surface opposite the first surface. The first electronic component is disposed on the first surface of the first substrate. The encapsulant encapsulates the first electronic component on the first surface of the first substrate. The feeding structure is disposed on the second surface of the first substrate without covering.

(5) In some embodiments, the present disclosure provides a semiconductor device package manufacturing method. The method includes providing a substrate having a first surface and a second surface opposite the first surface. The method also includes disposing a first semiconductor device on the first surface of the substrate. The method further includes encapsulating the first semiconductor device on the first surface of the substrate. The method also includes disposing a feeding structure on the second surface of the substrate. The method further includes exposing the feeding structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Aspects of some embodiments of the present disclosure are readily understood from the following detailed description when read with the accompanying figures. It is noted that various structures may not be drawn to scale, and dimensions of the various structures may be arbitrarily increased or reduced for clarity of discussion.

(2) FIG. 1 illustrates a layout of a semiconductor device package in accordance with some embodiments of the present disclosure.

(3) FIG. 2A illustrates a cross-sectional view of line A-A' of the semiconductor device package shown in FIG. 1.

(4) FIG. 2B illustrates a cross-sectional view of line A-A' of the semiconductor device package shown in FIG. 1.

(5) FIG. 2C illustrates a cross-sectional view of line A-A' of the semiconductor device package shown in FIG. 1.

(6) FIG. 2D, FIG. 2E, and FIG. 2F illustrate a method of manufacturing a semiconductor device package in accordance with some embodiments of the present disclosure.

(7) FIG. 3 illustrates application of the semiconductor device package shown in FIG. 1 in

accordance with some embodiments of the present disclosure.

(8) FIG. 4A illustrates an exploded view of an acoustic device in accordance with some embodiments of the present disclosure.

(9) FIG. 4B illustrates a perspective view of an assembly of the acoustic device as shown in FIG. 4A.

(10) FIG. 5A illustrates an exploded view of electronic components of an acoustic device in accordance with some embodiments of the present disclosure.

(11) FIG. 5B illustrates an assembly of the electronic components of the acoustic device in accordance with some embodiments of the present disclosure.

(12) FIG. 5C illustrates a side view of the assembly of the electronic components of the acoustic device in accordance with some embodiments of the present disclosure.

(13) FIG. 5D illustrates an exploded view of an acoustic device in accordance with some embodiments of the present disclosure.

(14) FIG. 5E illustrates a side view of an assembly of the acoustic device as shown in FIG. 5D.

(15) FIG. 6A illustrates an acoustic device in accordance with some embodiments of the present disclosure plugged in a human ear.

(16) FIG. 6B illustrates an acoustic device in accordance with some other embodiments of the present disclosure plugged in a human ear.

(17) FIG. 6C illustrates the acoustic device shown in FIG. 6B falls out of the human ear.

DETAILED DESCRIPTION

(18) FIG. 1 illustrates a layout of a semiconductor device package **1**. The semiconductor device package **1** can have a substrate **20**, a microphone **10**, a controller **11**, a sensor **12**, a connector **13**, a feeding structure **14**, ground pads **15** and electronic components **16**.

(19) The microphone **10**, controller **11**, sensor **12**, connector **13**, feeding structure **14**, ground pads **15** and electronic components **16** can be formed on the same side of the semiconductor device package **1**.

(20) The microphone **10** is exposed on the substrate **20**. The controller **11** is exposed on the substrate **20**. The sensor **12** is exposed on the substrate **20**. The connector **13** is exposed on the substrate **20**. The feeding structure **14** is exposed on the substrate **20**. The ground pads **15** are exposed on the substrate **20**. The electronic components **16** are exposed on the substrate **20**. A conductive pad **17** is disposed on the substrate **20**. Conductive pads **18** are disposed on the substrate **20**. The conductive pad **17** is exposed on the substrate **20**. The conductive pads **18** are exposed on the substrate **20**. The conductive pad **17** can be configured or used to connect the battery and the semiconductor device package **1**. The conductive pad **18** can be configured or used to connect the speaker and the semiconductor device package **1**. The semiconductor device package **1** can include a printed circuit board assembly. Semiconductor devices or semiconductor device packages can be mounted on the semiconductor device package **1**. The semiconductor device package **1** can be referred to as a system on module (SOM).

(21) A line A-A' may pass center of the semiconductor device package **1**. The line A-A' may pass center of a surface (e.g. top surface, not denoted in FIG. 1) of the substrate **20**. The line A-A' may pass the microphone **10** of the substrate **20**. The line A-A' may pass the feeding structure **14** of the substrate **20**.

(22) A line B-B' may pass center of the semiconductor device package **1**. The line B-B' may pass center of the surface of the substrate **20**. The line B-B' may pass the microphone **10** of the substrate **20**. As shown in FIG. 1, the microphone **10** can be disposed adjacent to the line A-A' of the surface of the substrate **20**. The microphone **10** can be disposed adjacent to the line B-B' of the surface of the substrate **20**. The feeding structure **14** may be disposed adjacent to the line A-A' of the surface of the substrate **20**.

(23) One of the benefits of the layout of the components of the semiconductor device package **1** may include an arrangement of the microphone **10** to be adjacent to or on the center of the top

surface (not denoted in FIG. 1) of the substrate **20**, which can fit or be used or be assembled in different housings (e.g. a housing for right ear and another housing for left ear).

(24) One of the benefits of the layout of the components of the semiconductor device package **1** may include an arrangement of the feeding structure **14** to be adjacent to or on the center of the top surface (not denoted in FIG. 1) of the substrate **20**, which can fit or be used or be assembled in different housings (e.g. a housing for right ear and another housing for left ear).

(25) FIG. 2A illustrates a cross-sectional view across line A-A' as shown in FIG. 1. A semiconductor device package **1a** can include a substrate **20**, an electronic component **21**, another electronic component **22**, an encapsulant **23** and a feeding structure **14**.

(26) The substrate **20** has a surface **20b** and a surface **20u** opposite the surface **20b**.

(27) The semiconductor device **21** can be disposed on the surface **20b** of the substrate **20**. The electronic component **22** can be disposed on the surface **20b** of the substrate **20**. The encapsulant **23** can encapsulate the electronic component **21** on the surface **20b** of the substrate **20**. The encapsulant **23** can encapsulate the electronic component **22** on the surface **20b** of the substrate **20**.

(28) The feeding structure **14** can be disposed on the surface **20u** of the substrate **20** and is devoid of encapsulant or molding material. The feeding structure **14** is exposed on the surface **20u** of the substrate **20**. The surface **20u** of the substrate **20** is devoid of encapsulant or molding material. Electronic components **25** are disposed on both surfaces **20b** and **20u** of the substrate **20**. The microphone **10** is disposed on the surface **20u**. The microphone **10** may include a semiconductor device package. The microphone **10** may include a microelectromechanical systems (MEMS) device. The microphone **10** may include a microelectromechanical systems (MEMS) device package. The microphone **10** is exposed. The sensor **12** is disposed on the surface **20u** of the substrate **20**. The connector **13** is disposed on the surface **20u** of the substrate **20**. The surface **20b** includes conductive pads for battery connection. The surface **20b** includes conductive pads for speaker driver connection. The feeding structure **14** has a surface **14u** higher than the electronic microphone **10**. The surface **14u** is higher than the sensor **12**. The surface **14u** is higher than the connector **13**. The feeding structure **14** is the highest among elements disposed on the surface **20u** of the substrate **20**.

(29) The electronic components **21** and **22** may include one or more of application processors, memories, digital signal processors, coders/decoders, power management integrated circuits, DC converters, low dropout outputs or filters. The encapsulant **23** may include different kinds of molding materials. The feeding structure **14** includes a spring or a conductive pin. The feeding structure **14** includes a POGO pin. The feeding structure **14** also includes an antioxidation layer. The feeding structure **14** is configured to and can be used to feed a radio frequency (RF) signal to an antenna. The electronic components **25** may include one or more of resistors, inductors or capacitors. The sensor **12** may include a G-sensor or other sensors. The connector **13** includes a zero insertion force connector or a board-to-board connector for peripheral function flexible printed circuit (FPC) connection.

(30) FIG. 2B illustrates another cross-sectional view of line A-A' of the semiconductor device package shown in FIG. 1. Electronic component **21'** is disposed on a surface **20'b** of a substrate **20'**. Electronic component **22'** is disposed on the surface **20'b** of the substrate **20'**. The surface **20'u** is opposite to the surface **20'b** of the substrate **20'**. Electronic components **25** are disposed on both surfaces **20'b** and **20'u** of the substrate **20'**. Encapsulant **23** surrounds the elements disposed on the surface **20'b** of the substrate **20'**. A substrate **20** has a surface **20b** and a surface **20u** opposite the surface **20b**. A feeding structure **14** is disposed on the surface **20u** of the substrate **20**. The feeding structure **14** is exposed. The surface **20u** of the substrate **20** is devoid of encapsulant or molding material. Electronic components **25** are disposed on both surfaces **20b** and **20u** of the substrate **20**. A microphone **10** is disposed on the surface **20u**. The microphone **10** is also exposed. A sensor **12** is disposed on the surface **20u** of the substrate **20**. A connector **13** is disposed on the surface **20u** of the substrate **20**. Encapsulant **23** surrounds the elements disposed between the surface **20b** of the

substrate **20** and the surface **20'u** of the substrate **20'**.

(31) FIG. 2C illustrates a further cross-sectional view of line A-A' of the semiconductor device package shown in FIG. 1. Electronic component **21''** is disposed on a surface **20''b** of a substrate **20''**. Electronic component **22''** is disposed on the surface **20''b** of the substrate **20''**. The surface **20''u** is opposite to the surface **20''b** of the substrate **20''**. Electronic components **25** are disposed on both surfaces **20''b** and **20''u** of the substrate **20''**. Encapsulant **23** surrounds the elements disposed on the surface **20''b** of the substrate **20''**. Electronic component **21'** is disposed on a surface **20'b** of a substrate **20'**. Electronic component **22'** is disposed on the surface **20'b** of the substrate **20'**. The surface **20'u** is opposite to the surface **20'b** of the substrate **20'**. Electronic components **25** are disposed on both surfaces **20'b** and **20'u** of the substrate **20'**. Encapsulant **23** surrounds the elements disposed on the surface **20'b** of the substrate **20'**. A substrate **20** has a surface **20b** and a surface **20u** opposite the surface **20b**. A feeding structure **14** is disposed on the surface **20u** of the substrate **20**. The feeding structure **14** is exposed. The surface **20u** of the substrate **20** is devoid of encapsulant or molding material. Electronic components **25** are disposed on both surfaces **20b** and **20u** of the substrate **20**. A microphone **10** is disposed on the surface **20u**. The microphone **10** is also exposed. A sensor **12** is disposed on the surface **20u** of the substrate **20**. A connector **13** is disposed on the surface **20u** of the substrate **20**. Encapsulant **23** surrounds the elements disposed between the surface **20b** of the substrate **20** and the surface **20'u** of the substrate **20'**. Encapsulant **23** also surrounds the elements disposed between the surface **20'b** of the substrate **20'** and the surface **20''u** of the substrate **20''**.

(32) FIGS. 2D, 2E and 2F illustrates a method for manufacturing the semiconductor device package **1a** shown in FIG. 2A.

(33) As shown in FIG. 2D, a substrate **20** having a surface **20b** and a surface **20u** opposite the surface **20b** is provided.

(34) As shown in FIG. 2E, an electronic component **21** is disposed on the surface **20b** of the substrate **20**. Another electronic component **22** is also disposed on the surface **20b** of the substrate **20**. Electronic components **25** are disposed on both of the surfaces **20b** and **20u** of the substrate **20**.
A

(35) As shown in FIG. 2F, the electronic component **21** is encapsulated on the surface **20b** of the substrate **20** by an encapsulant **23**. Another electronic component **22** is encapsulated on the surface **20b** of the substrate **20** by the encapsulant **23**. Electronic components **25** are encapsulated on the surface **20b** of the substrate **20** by the encapsulant **23**. A feeding structure **14** is disposed on the surface **20u** of the substrate **20**. The feeding structure **14** is exposed. A microphone **10** is disposed on the surface **20u** of the substrate **20**. The microphone **10** is exposed. The surface **20u** of the substrate **20** is also exposed. Electronic components **25** are disposed on the surface **20u** of the substrate **20**. A sensor **12** is disposed on the surface **20u** of the substrate **20**. A connector **13** is disposed on the surface **20u** of the substrate **20** to form the semiconductor device package as shown in FIG. 2A.

(36) FIG. 3 illustrates application of semiconductor device packages of earphones. Acoustic device (e.g. earphones) **30R** is plugged into human ears **31R**. Acoustic device **30L** is plugged into human ears **31L**. The right acoustic device **30R** includes semiconductor device packages **32R**. The left acoustic device **30L** includes semiconductor device packages **32L**. The layout of the semiconductor device package **32R** is the same or identical to the layout of the semiconductor device package **32L**, except that the semiconductor device package **32R** is rotated by substantially 180 degrees with respect to the semiconductor device package **32L** after assembly. Accordingly, one layout of semiconductor device package is used for both acoustic devices **30R** and **30L**, which can reduce cost. The acoustic devices **30R** and **30L** may include the semiconductor device packages **1a**, **1b** or **1c** shown in FIGS. 1A, 1B and 1C.

(37) FIG. 4A illustrates an exploded view of an acoustic device **4** in accordance with some embodiments of the present disclosure. The acoustic device **4** includes a speaker **40**, a battery **41**, a

semiconductor device package 42, an interconnect structure 43 and a housing 44. The speaker has a surface 40a and a surface 40b opposite to the surface 40a. The battery 41 has a surface 41a, a surface 41b opposite the surface 41a, and a surface 41c extended between the surface 41a and the surface 41b. The speaker 40 is disposed adjacent to the surface 41a of the battery 41. The surface 40a of the speaker 40 faces with the surface 41a of the battery 41. The structure of the semiconductor device package 42 is illustrated in the cross-sectional view of the semiconductor device package 1a, 1b or 1c in FIGS. 2A-2C, respectively. The speaker 40 is connected to the semiconductor device package 42 by wires 45. The wires 45 do not penetrate the battery 41. The speaker 40 is separated from the battery 41 by a space S1. There is no turbulent flow in the space S1. The space S1 can influence the acoustic performance of the acoustic device 4. The housing 44 encloses the speaker 40, the battery 41, the semiconductor device package 42, and the interconnect structure 43. The interconnect structure 43 connects the battery 41 and the semiconductor device package 42. The interconnect structure 43 includes flexible print circuit (FPCs) or other interconnect structures.

(38) FIG. 4B illustrates a perspective view of an assembly of the acoustic device 4 as shown in FIG. 4A. The semiconductor device package 42 is disposed adjacent to the surface 41c of the battery 41. The semiconductor device package 42 is electrically connected to the surface 41c of the battery 41 by the interconnect structure 43. The speaker 40 is separated from the surface 41a of the battery 41 by the space S1. The space S1 is devoid of electronic components. The space S1 may have a plate for separating the speaker and the battery 41. The space S1 may have conductive wires therein. The space S1 may have solely conductive wires therein. The space S1 may have no active components. The space S1 may have no passive components.

(39) The semiconductor device package 42 is disposed between the housing 44 and the surface 41c of the battery 41. The speaker 40 has a surface 40a facing the surface 41a of the battery 41. The semiconductor device package 42 is disposed between the housing 44, the surface 41c of the battery 41 and the surface 40a of the speaker 40. The semiconductor device package 42 is surrounded by the housing 44, the surface 41c of the battery 41 and the surface 40a of the speaker 40. The semiconductor device package 42 is disposed in a space defined by the housing 44, the surface 41c of the battery 41 and the speaker 40. The housing 44 encloses the speaker 40, the battery 41 and the semiconductor device package 42. The housing 44 has an end 44a and an end 44b opposite the end 44a. Antenna patterns 46 are formed on the housing 44 to work as antennas. The feeding structure 14 may be in contact with the antenna patterns 46 of the housing 44 of the acoustic device 4.

(40) Referring again to FIG. 4B, assuming the speaker 40 can have a width or diameter of approximately 6 mm, a maximum distance Z1 between the end 44a (which can be fitted or plugged into an ear) and the end 44b along the z axis can be equal to or less than 23 mm. For example, assuming the speaker 40 can have a width or diameter of approximately 6 mm, a distance Z1 between the end 44a (which can be fitted or plugged into an ear) and the end 44b along the z axis can be equal to or less than 20 mm. Accordingly, the semiconductor device package 42 can facilitate miniaturization of the acoustic device 4. Arrangement of the semiconductor device package 42 can facilitate miniaturization of the acoustic device 4.

(41) For example, assuming the speaker 40 can have a width or diameter of approximately 10 mm, a distance Z1 between the end 44a (which can be fitted or plugged into an ear) and the end 44b along the z axis can be equal to approximately 23 mm.

(42) FIG. 5A illustrates an exploded view of electronic components of an acoustic device. Circuit boards 51, 52 and 53 of an acoustic device are connected by an interconnect structure 54. Electronic components are formed on the circuit boards 51, 52 and 53. The interconnect structure 54 may include FPCs or other interconnect structures. A battery 55 can be provided in the acoustic device.

(43) The acoustic device, which includes circuit boards 51, 52 and 53, can be fitted or plugged into

a left ear. However, another acoustic device to be fitted or plugged into a right ear may include circuit boards each having layout different from circuit boards **51**, **52** and **53**, which may inevitably increase the cost.

(44) FIG. **5B** illustrates an assembly of the electronic components of the acoustic device in accordance with some embodiments of the present disclosure. The circuit boards **51**, **52** and **53** are stacked. The thickness of the stack is **T1**. The circuit boards **51** and **52** and the interconnect structure **54** surround a battery **55**. The circuit boards **52** and **53** and the interconnect structure **54** surround a mechanical component **56**.

(45) FIG. **5C** illustrates a side view of the assembly of the electronic components of the acoustic device. FIG. **5C** shows that the circuit board **51** can be above the battery **55**. The circuit board **52** can be under the battery **55**. The circuit boards **51** and **52** and the interconnect structure **54** surround the battery **55**. The circuit board **53** is separated from the battery **55**. As shown in FIG. **5C**, the mechanical component **56** may be ignored.

(46) FIG. **5D** illustrates an exploded view of an acoustic device **5** including the stack shown in FIG. **5C**. The acoustic device **5** may include a speaker **50**, circuit boards **51**, **52** and **53**, the interconnect structure **54**, a battery **55** and a housing **57**. The circuit board **53** is disposed between the speaker **50** and the battery **55**, which causes electromagnetic interference to the speaker **50** and thus degrades the acoustic performance of the acoustic device **5**. The housing **57** encloses the speaker **50**, the circuit boards **51**, **52** and **53**, the interconnect structures **54**, and the battery **55**. Antenna patterns **58** are formed on the housing **57**. A feeding point (not denoted in FIG. **5D**) on the circuit board **53** is in contact with the antenna patterns **58**.

(47) FIG. **5E** illustrates a side view of an assembly of the acoustic device **5** shown in FIG. **5D**. In some embodiments, assuming the diameter of the speaker inside the acoustic device **5** is 6 mm, the maximum distance **Z2** between an end **57a** (which can be fitted or plugged into an ear) and an end **57b** is equal to or larger than 23 mm. Because the thickness **T1** shown in FIG. **5B** is larger than the distance between an end **42a** of the semiconductor device package **42** and the end **41a** of the battery **41** along the *z* axis in FIG. **4B**, the maximum distance **Z2** of the acoustic device **5** is larger than the maximum distance **Z1** of the acoustic device **4**.

(48) Referring to FIG. **6A**, the acoustic device **4** shown in FIG. **4B** is plugged into a human ear. The center of gravity **CM1** may be adjacent or close to the end **44b**. Due to the miniaturization of the acoustic device **4**, when plugging the acoustic device **4** into the ear, the acoustic device **4** can be in contact with a point **P** of the ear. Accordingly, the acoustic device **4** can have a relatively good support of the ear. Accordingly, the acoustic device **4** would not easily fall out of the even when the person wearing the acoustic device **4** is moving or sweating.

(49) Referring to FIG. **6B**, the acoustic device **5** shown in FIG. **5E** is plugged into a human ear. The center of gravity **CM2** may be adjacent or close to the end **57b**.

(50) The acoustic device **5**, which may have a relatively great size, may have a major portion (except the plug) that is floating. The acoustic device **5** may be spaced from the point **P** of the human ear. A distance **D1** is between the top surface of the acoustic device **5** and the human ear.

(51) FIG. **6C** illustrates that the acoustic device **5** starts falling off the ear when the person wearing the acoustic device **5** swears, moves or change positions. The falling of the acoustic device **5** may be resulted from an unstable state as shown in FIG. **6B**, in which the center of gravity **CM2** may cause the acoustic device **5**, which lacks support of the ear, relatively easy to fall out of the ear.

(52) It is contemplated that a distance **D2** from the top surface of the acoustic device **5** and the human ear is increasing due to the falling of the acoustic device **5**. The distance **D2** is larger than the distance **D1**.

(53) Spatial descriptions, such as “above,” “below,” “up,” “left,” “right,” “down,” “top,” “bottom,” “vertical,” “horizontal,” “side,” “higher,” “lower,” “upper,” “over,” “under,” and so forth, are indicated with respect to the orientation shown in the figures unless otherwise specified. It should be understood that the spatial descriptions used herein are for purposes of illustration only, and that

practical implementations of the structures described herein can be spatially arranged in any orientation or manner, provided that the merits of embodiments of this disclosure are not deviated from by such an arrangement.

(54) As used herein, the terms “approximately,” “substantially,” “substantial” and “about” are used to describe and account for small variations. When used in conjunction with an event or circumstance, the terms can refer to instances in which the event or circumstance occurs precisely as well as instances in which the event or circumstance occurs to a close approximation. For example, when used in conjunction with a numerical value, the terms can refer to a range of variation less than or equal to $\pm 10\%$ of that numerical value, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to $\pm 0.5\%$, less than or equal to $\pm 0.1\%$, or less than or equal to $\pm 0.05\%$. For example, a first numerical value can be deemed to be “substantially” the same or equal to a second numerical value if the first numerical value is within a range of variation of less than or equal to $\pm 10\%$ of the second numerical value, such as less than or equal to $\pm 5\%$, less than or equal to $\pm 4\%$, less than or equal to $\pm 3\%$, less than or equal to $\pm 2\%$, less than or equal to $\pm 1\%$, less than or equal to $\pm 0.5\%$, less than or equal to $\pm 0.1\%$, or less than or equal to $\pm 0.05\%$. For example, “substantially” perpendicular can refer to a range of angular variation relative to 90° that is less than or equal to $\pm 10^\circ$, such as less than or equal to $\pm 5^\circ$, less than or equal to $\pm 4^\circ$, less than or equal to $\pm 3^\circ$, less than or equal to $\pm 2^\circ$, less than or equal to $\pm 1^\circ$, less than or equal to $\pm 0.5^\circ$, less than or equal to $\pm 0.1^\circ$, or less than or equal to $\pm 0.05^\circ$.

(55) Two surfaces can be deemed to be coplanar or substantially coplanar if a displacement between the two surfaces is no greater than $5\text{ }\mu\text{m}$, no greater than $2\text{ }\mu\text{m}$, no greater than $1\text{ }\mu\text{m}$, or no greater than $0.5\text{ }\mu\text{m}$. A surface can be deemed to be substantially flat if a displacement between a highest point and a lowest point of the surface is no greater than $5\text{ }\mu\text{m}$, no greater than $2\text{ }\mu\text{m}$, no greater than $1\text{ }\mu\text{m}$, or no greater than $0.5\text{ }\mu\text{m}$.

(56) As used herein, the singular terms “a,” “an,” and “the” may include plural referents unless the context clearly dictates otherwise.

(57) As used herein, the terms “conductive,” “electrically conductive” and “electrical conductivity” refer to an ability to transport an electric current. Electrically conductive materials typically indicate those materials that exhibit little or no opposition to the flow of an electric current. One measure of electrical conductivity is Siemens per meter (S/m). Typically, an electrically conductive material is one having a conductivity greater than approximately 10^4 S/m , such as at least 10^5 S/m or at least 10^6 S/m . The electrical conductivity of a material can sometimes vary with temperature. Unless otherwise specified, the electrical conductivity of a material is measured at room temperature.

(58) Additionally, amounts, ratios, and other numerical values are sometimes presented herein in a range format. It is to be understood that such range format is used for convenience and brevity and should be understood flexibly to include numerical values explicitly specified as limits of a range, but also to include all individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly specified.

(59) While the present disclosure has been described and illustrated with reference to specific embodiments thereof, these descriptions and illustrations are not limiting. It should be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the true spirit and scope of the present disclosure as defined by the appended claims. The illustrations may not be necessarily drawn to scale. There may be distinctions between the artistic renditions in the present disclosure and the actual apparatus due to manufacturing processes and tolerances. There may be other embodiments of the present disclosure which are not specifically illustrated. The specification and drawings are to be regarded as illustrative rather than restrictive. Modifications may be made to adapt a particular situation, material, composition of matter, method, or process to the objective, spirit and scope of the present

disclosure. All such modifications are intended to be within the scope of the claims appended hereto. While the methods disclosed herein have been described with reference to particular operations performed in a particular order, it will be understood that these operations may be combined, sub-divided, or re-ordered to form an equivalent method without departing from the teachings of the present disclosure. Accordingly, unless specifically indicated herein, the order and grouping of the operations are not limitations of the present disclosure.

(60) As used herein, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” “lower,” “left,” “right” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly. It should be understood that when an element is referred to as being “connected to” or “coupled to” another element, it may be directly connected to or coupled to the other element, or intervening elements may be present.

(61) The foregoing outlines features of several embodiments and detailed aspects of the present disclosure. The embodiments described in the present disclosure may be readily used as a basis for designing or modifying other processes and structures for carrying out the same or similar purposes and/or achieving the same or similar advantages of the embodiments introduced herein. Such equivalent constructions do not depart from the spirit and scope of the present disclosure, and various changes, substitutions, and alterations may be made without departing from the spirit and scope of the present disclosure.

Claims

1. An electronic device, comprising: a substrate; a connector disposed over the substrate; a feeding structure disposed over the substrate and configured to feed a radio frequency (RF) signal to an antenna; and a conductive pad disposed adjacent to the connector and configured to connect a speaker, wherein a central axis of a side of a surface of the substrate in a plan view passes the feeding structure.
2. The electronic device of claim 1, wherein the feeding structure is in contact with the antenna.
3. The electronic device of claim 2, further comprising a housing for accommodating the substrate, the connector and the feeding structure, wherein the antenna is formed on the housing.
4. The electronic device of claim 1, wherein the substrate is electrically connected to the speaker through the connector and a flexible printed circuit, wherein the flexible printed circuit connects the connector with the speaker.
5. The electronic device of claim 1, further comprising: an electronic component disposed under the substrate and electrically connected to the substrate; and an encapsulant encapsulating the electronic component and in contact with the substrate.
6. The electronic device of claim 1, further comprising a microphone and a sensor disposed between the connector and the feeding structure in the plan view.
7. The electronic device of claim 1, wherein the conductive pad is disposed between the connector and the feeding structure in the plan view.
8. The electronic device of claim 1, wherein the conductive pad is disposed outside a projection of the feeding structure toward the connector in the plan view.
9. An electronic device, comprising: a housing; a battery accommodated by the housing; and a device comprising: a substrate; a connector disposed over the substrate; a conductive pad disposed adjacent to the connector and configured to connect the battery; and a feeding structure disposed over the substrate and configured to feed a radio frequency (RF) signal to an antenna, wherein a height of the feeding structure is greater than a height of the connector.

10. The electronic device of claim 9, further comprising a flexible printed circuit connecting the connector with the battery.
 11. The electronic device of claim 10, wherein a longitudinal direction of the flexible printed circuit is not perpendicular to a longitudinal direction of a lateral surface of the battery.
 12. The electronic device of claim 10, wherein a longitudinal direction of the flexible printed circuit is non-parallel to a central axis of the battery.
 13. The electronic device of claim 9, wherein the housing includes an antenna pattern, wherein the device is disposed between the antenna pattern and the battery, wherein the device is disposed outside a projection of the battery along an axial direction of the battery.
 14. The electronic device of claim 13, wherein the device includes a feeding structure disposed on a first surface of the substrate and in contact with the antenna pattern, wherein the substrate has a second surface opposite to the first surface and facing the battery.
 15. The electronic device of claim 9, further comprising an encapsulant encapsulating the substrate, wherein the substrate and the connector are separated from the battery by the encapsulant.
 16. The electronic device of claim 9, further comprising a speaker accommodated by the housing and separated from the battery by a space, wherein the battery has a first surface facing the speaker and a second surface opposite to the first surface, wherein an extension of the second surface of the battery passes the device.
 17. An electronic device, comprising: a housing; a speaker accommodated by the housing; a battery accommodated by the housing, wherein the battery has a first surface facing the speaker, a second surface opposite to the first surface, and a lateral surface extending between the first surface and the second surface; a device accommodated by the housing and comprising: a substrate having a first surface and a second surface opposite to the first surface; and an encapsulant encapsulating the first surface of the substrate; and a flexible printed circuit disposed over the second surface of the substrate and electrically connecting the substrate to the battery and the speaker, wherein the flexible printed circuit is electrically connected to the lateral surface of the battery.
 18. The electronic device of claim 17, further comprising a connector disposed on the second surface of the substrate and connecting the substrate with the flexible printed circuit.
 19. The electronic device of claim 17, wherein the device further comprises a first electronic component disposed under the first surface of the substrate and encapsulated by the encapsulant, wherein the first electronic component is spaced apart from the battery by the encapsulant.
 20. The electronic device of claim 17, wherein the lateral surface of the battery includes a first portion and a second portion opposite to the first portion, wherein a bottom surface of the device faces the first portion of the lateral surface of the battery, wherein the flexible printed circuit extends from a top surface of the device to the second portion of the lateral surface of the battery.
-